
EMBEDDED HACKING

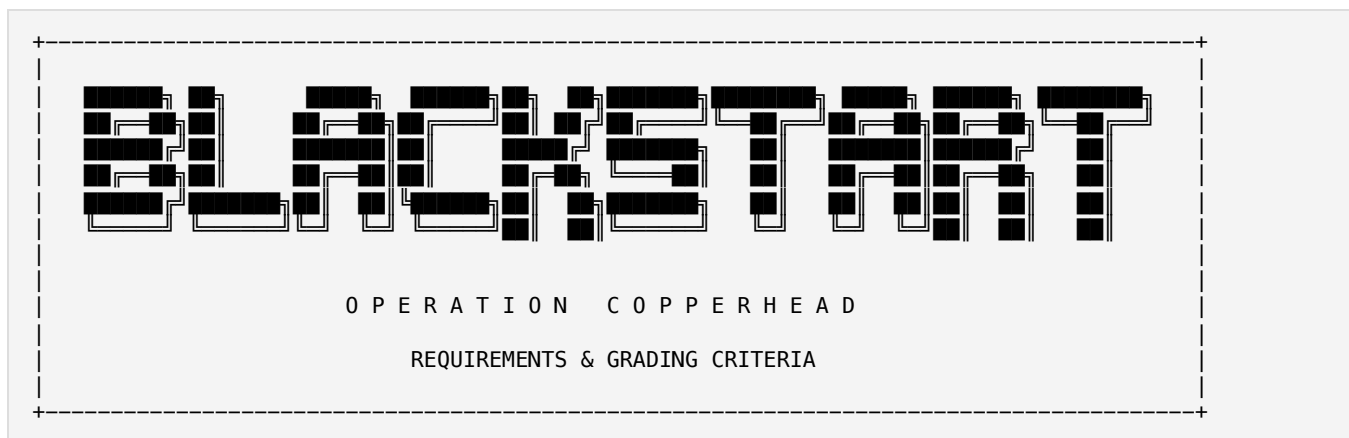
Reverse Engineering the Raspberry Pi Pico 2

Operation Copperhead - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

September 19, 2026

For educational and defensive security research only



Project Overview

Students are the reverse-engineering reserve team called in after DEEPLINE Metro Authority's rebuilt DEEPLINE-AUTH relay image shipped four corrupted engineering constants: a miscalibrated release threshold, a false TRACK banner string, an overstated block length, and a poisoned ARX signal seed. Students reverse engineer `CTF-02.bin` with Ghidra, patch all four defects, capture the runtime-derived signal key live in GDB, recover the Ouroboros authority frame, export a corrected image, flash it to real hardware, and prove the corrected behavior on a physical Pico 2.

The challenge is a standalone capstone exercise and contains no answer, constant, address, bug, or patch belonging to any other course assignment.

Learning Objectives

- Decode an ARM Cortex-M33 vector and boot table and identify the reset handler and initial stack pointer.
- Translate Thumb reset-vector addresses into real function entry points and trace literal-pool entries to their data.
- Locate four corrupted constants: a boundary comparison, a status string, an 8-byte IEEE-754 double, and an ARX signal seed.
- Capture a runtime-derived key with GDB, override a register, and set a watchpoint on stored SRAM state.
- Recover and authenticate an Argon2id plus XChaCha20-Poly1305 authority frame.

Students must use only Weeks 1-8 concepts: ARM registers, stack behavior, USB-CDC output, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, Thumb addressing, data segments and literal pools, condition-code analysis, runtime key derivation, and the Argon2id plus XChaCha20-Poly1305 authenticated gate.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot	PNG/JPG	1.1
2	Vector table and boot table	Inside <code>CTF-02-Answers.md</code>	1.2

#	Deliverable	Format	Criterion
3	<code>main()</code> and status-loop table	Inside <code>CTF-02-Answers.md</code>	1.3
4	Literal pool trace	Inside <code>CTF-02-Answers.md</code>	1.4
5	Bug #1 evidence and patches	Inside <code>CTF-02-Answers.md</code>	2.1-2.4
6	Bug #2 evidence and patch	Inside <code>CTF-02-Answers.md</code>	3.1-3.4
7	Bug #3 evidence and patch	Inside <code>CTF-02-Answers.md</code>	4.1-4.3
8	Bug #4 evidence and patch	Inside <code>CTF-02-Answers.md</code>	5.1-5.4
9	GDB register capture	Inside <code>CTF-02-Answers.md</code>	6.1-6.4
10	Ouroboros gate recovery and auth	Inside <code>CTF-02-Answers.md</code>	7.1-7.4
11	<code>CTF-02_fixed.bin</code>	BIN file	8.1
12	<code>CTF-02_fixed.uf2</code>	UF2 file	8.2
13	Corrected console transcript	Inside <code>CTF-02-Answers.md</code>	8.3
14	Summary table of all patches	Inside <code>CTF-02-Answers.md</code>	8.4
15	Written reflection	Inside <code>CTF-02-Answers.md</code>	9.1-9.2

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target
USB-CDC virtual serial console	Observe output and type the gate passphrase
SWD debug probe	GDB inspection
Ghidra	Static analysis and binary patching
GDB	Dynamic analysis and register capture
Python (<code>uf2conv.py</code>)	UF2 conversion
<code>CTF-02.bin</code> and <code>CTF-02.uf2</code>	Supplied artifacts

Console settings: **USB-CDC virtual COM port, 115200 baud, 8 data bits, no parity, 1 stop bit.**

Artifact Identity

The instructor-issued artifact hashes are:

```
CTF-02.bin  85330C37CD0897746B1AF447E4BAC371DDE2042ABD2D61D58A61FE2A8EEF3537
CTF-02.uf2  F3CD4840260DB820D792758CECACC5297BEF1971B9EACF7601279256D8AF1EAB
```

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (12 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 1.1: Ghidra Project Setup	3	Correct project name, ARM Cortex 32-bit little endian, base 0x10000000	One item off	Not set up
Criterion 1.2: Vector Table Decoding	3	Correct base, initial SP, reset pointer	One missing	Not found
Criterion 1.3: main() and Status-Loop Addresses	4	Both addresses correct	One correct	Neither found
Criterion 1.4: Thumb Addressing and Literal Pool	2	Bit 0 cleared and one pool entry traced to its string	Partial	Incorrect

Task 2: Find and Patch Bug #1: The Miscalibrated Release Threshold (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 2.1: Locate Compare Sites A and B	6	Both addresses and original bytes	One site	Not found
Criterion 2.2: Correct Immediate-Value Reasoning	4	Explains the < to <= transform and gives 0x3B	Correct value, no reasoning	Wrong value
Criterion 2.3: Patch Compare Sites A and B	4	Both byte changes verified	One site	Not patched
Criterion 2.4: Explain Why Both Sites Must Be Patched	1	Clear explanation of the two independent comparisons	Vague	Missing

Task 3: Find and Patch Bug #2: The False TRACK Banner (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 3.1: Locate the Banner String	3	Correct address and cross-reference	Approximate	Not found
Criterion 3.2: Patch Six Characters	4	All six bytes changed, length preserved	Correct text, wrong bytes documented	Wrong length
Criterion 3.3: Character-by-Character Documentation	2	Original vs patched byte for all six characters	Partial	Missing
Criterion 3.4: Explain the Danger of a Hardcoded Status Word	1	Clear, specific reasoning	Generic	Missing

Task 4: Find and Patch Bug #3: The Block Length Constant (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 4.1: Locate the .data Double	3	Correct address traced from the BLOCK LENGTH print	Approximate	Not found
Criterion 4.2: IEEE-754 Bytes and Print Math	4	Original and patched 8-byte double with print math (3200 M to 320 M)	Correct patch, no math	Wrong bytes
Criterion 4.3: Patch to Print 320 M	3	Console shows <code>BLOCK LENGTH: 320 M</code>	Wrong bytes	Not patched

Task 5: Find and Patch Bug #4: The Signal Seed (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 5.1: Locate SIGNAL_SPEC and the Seed	5	<code>0x2D879291</code> located and wrong seed <code>0x0A0A0A0A</code> found	Partial	Not found
Criterion 5.2: Patch the Seed	4	Seed bytes changed to <code>74 65 20 6B</code>	Wrong byte	Not patched
Criterion 5.3: Explain the ARX Derivation	3	Correct trace of the per-cycle derivation	Vague	Missing
Criterion 5.4: Separate the Security Layers	3	Correctly explains what the seed fixes versus the gate	Generic	Missing

Task 6: GDB Register Capture of the Derived Key (15 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 6.1: Breakpoint at the Derive Return	4	Correct address and <code>\$r0</code> read as the bug-derived key	Address off	Not found
Criterion 6.2: Inspect the Two Arguments	4	Live seed and derived IV captured at the second call	One correct	Missing
Criterion 6.3: Override the Register	4	<code>\$r0</code> set to <code>0x2D879291</code> and the next cycle shows <code>OK</code>	Partial	Missing
Criterion 6.4: Watchpoint on the Stored Key	3	Watchpoint on the SRAM key location documented	Approximate	Missing

Task 7: Recover the Ouroboros Authority Frame (10 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 7.1: Locate Salt, Nonce, Ciphertext, and Tag	4	All three addresses correct in flash	Two correct	Not found

Criterion	Points	Full credit	Partial credit	No credit
Criterion 7.2: Document Argon2id Parameters and Payload Contract	2	Correct memory/time/parallelism and payload layout	Partial	Missing
Criterion 7.3: Authenticate with the 12-Word Passphrase	2	AUTHORITY FRAME: VERIFIED , LED on, payload printed	Partial	Not shown
Criterion 7.4: State the Honest Quantum Boundary	2	Grover halves symmetric exponents; not strict PQC	Generic	Misstates

Task 8: Export and Verify (8 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 8.1: Export CTF-02_fixed.bin	1	Valid patched binary	Corrupted	Not submitted
Criterion 8.2: Convert to CTF-02_fixed.uf2	1	Correct base and family flags	Wrong flags	Not submitted
Criterion 8.3: Hardware Verification	4	Corrected console output confirmed on hardware	Some lines corrected	No verification
Criterion 8.4: Summary Table of All Patches	2	Complete address and before/after table	Missing entries	No table

Task 9: Written Reflection (5 points)

Criterion	Points	Full credit	Partial credit	No credit
Criterion 9.1: "Rushed Build" Is Not an Excuse	2	Specific, grounded reasoning	Generic	Missing
Criterion 9.2: One Engineering Practice per Failure Area	3	Concrete practices for the bugs and for image authenticity	Names some	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Patching only one threshold site	One status line still lies	Patch both 0x10000302 and 0x10000312
Assuming the immediate equals the limit	Off-by-one, wrong boundary	Use 0x3B (59), not 0x3C (60)
Replacing a string with a different length	Corrupts adjacent flash	NORMAL and DANGER are both 6 bytes
Treating the block length as an integer	Misses the 8-byte double	Follow the value into .data , decode IEEE-754
Using the wrong 0.32 bytes	Prints 316 M instead of 320 M	Use 7B 14 AE 47 E1 7A D4 3F
Treating an odd vector address as invalid	Thumb analysis fails	Clear bit 0

Pitfall	Consequence	Avoidance
Starting the seed patch at the wrong offset	Wrong seed, key never matches	Seed is at <code>0x1000EC70</code>

How To Breadboard

- **Raspberry Pi Pico 2** powered over USB.
- **USB-CDC virtual serial console:** open the Pico's COM port at 115200 baud, 8 data bits, no parity, 1 stop bit.
- **SWD debug probe:** connect SWCLK, SWDIO, GND, and 3.3 V to the Pico debug header for GDB inspection and register capture.
- No other peripherals are required; the authority LED is on-board.

Memory Map Reference

Region	Address	Purpose
Bootrom	<code>0x00000000</code>	Immutable boot code
Flash/XIP	<code>0x10000000</code>	Vector table, code, rodata, <code>.data</code> init image
SRAM	<code>0x20000000</code>	Stack and writable state

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `CTF-02_fixed.bin`, and `CTF-02_fixed.uf2`.
- Write all written answers in a single file named `CTF-02-Answers.md` inside that folder.
- ZIP the folder as `lastname-firstname-CTF-02.zip`.
- Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.

Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79

Grade	Percentage	Points
F	0-69%	0-69

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational railway, metro system, public network, military system, or third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, keys, passphrases, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Week 1
USB-CDC output and console capture	Week 2
Vector tables, reset startup, and XIP	Week 2
Ghidra static analysis and binary patching	Week 3
Data segments, literal pools, IEEE-754	Week 4
Condition-code analysis	Week 5
Runtime key derivation and GDB register capture	Week 6
Argon2id and XChaCha20-Poly1305 authenticated gate	Week 8